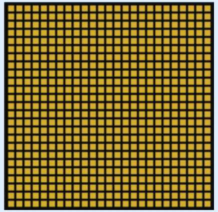

Gigantic MIMO:
Next-Generation Antenna Technology?

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A special thanks to

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Parisa Ramezani

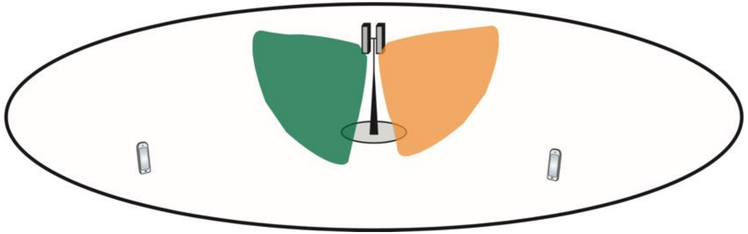
Alva Kosasih

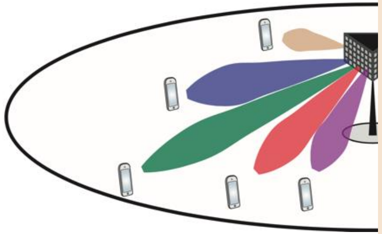
Ferdi Kara

Evolution of Multiple Access in Cellular Networks

1. Sectorization and orthogonal access

2. Multi-user MIMO (multiple-input multiple-output)



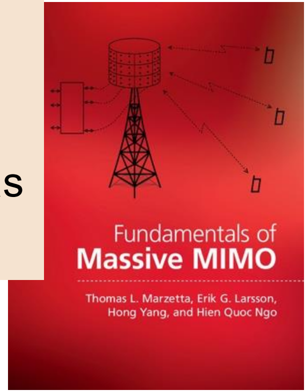


Assumption

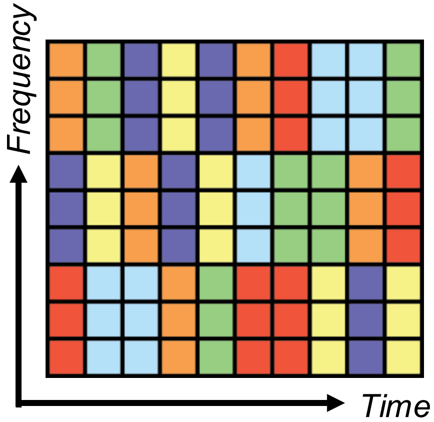
$M \gg K$

M = #antennas

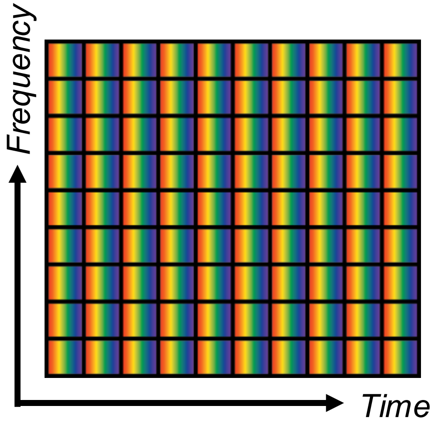
K = #users



Time-frequency scheduling



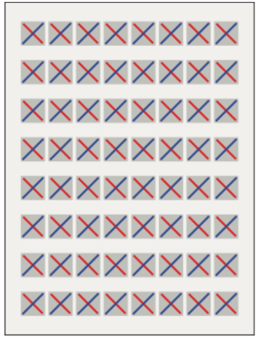
User-separation by beamforming



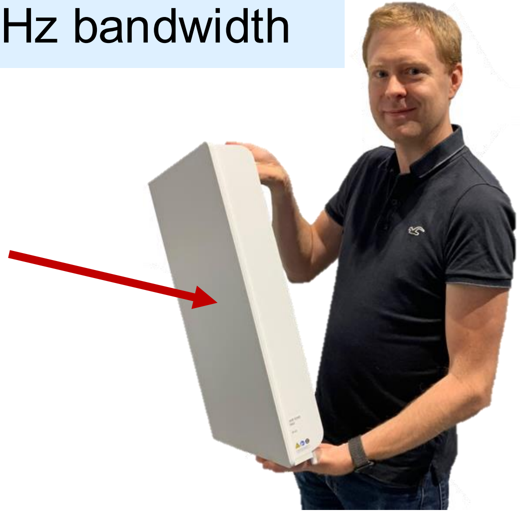
Massive MIMO in 5G

n78 band: 3.3-3.8 GHz

100 MHz bandwidth



64 antennas



One device: 710 Mbps (average)

MU-MIMO with 8 user devices

- 16 spatial layers, 2 layers/user
- 650 Mbps per device (average)
- 5.2 Gbps in total (52 bit/s/Hz)

$$\frac{M}{K} = 4$$

4 antennas

Reference: Signals Research Group, "The Industry's First Independent Benchmark Study of 5G NR MU-MIMO," 2020

6G Networks in the **Upper Mid-Band** (7-24 GHz)

6G Candidate bands

- 4.4-4.8 GHz (400 MHz)
- 7.1-8.4 GHz (650-1275 MHz)
- 14.8-15.35 GHz (550 MHz)

Current 5G band

- 3.3-3.8 GHz (400 MHz)

\+ Legacy bands (e.g., 0.7, 2, 2.6 GHz)

6G will only have slightly more bandwidth than 5G!

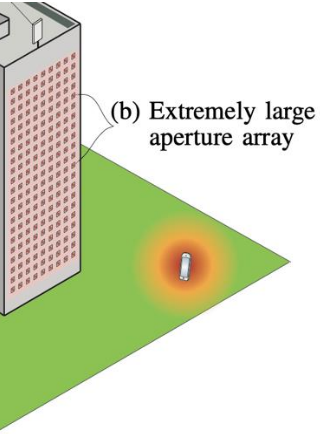
Per operator and base station: **4G** 20 MHz $\xrightarrow{5\times}$ **5G** 100 MHz $\xrightarrow{2\times}$ **6G** 200 MHz

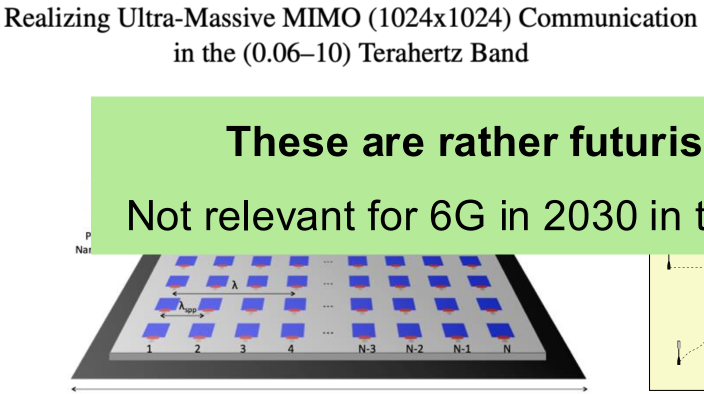
Can “better MIMO” be the distinguishing 6G factor?

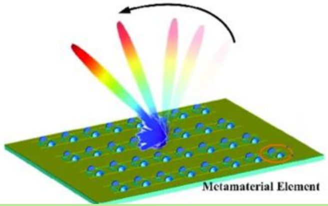
Many New “6G” MIMO Terms

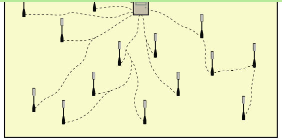
- Ultra Massive MIMO: mmWave and THz systems
- XL-MIMO, Extremely Large Aperture Arrays: Physically Large Surfaces
- Cell-free Massive MIMO: Cooperation between small arrays
- Holographic MIMO, dynamic metasurfaces Tiny antenna spacings

Near-field effects highly prominent









These are rather futuristic concepts!

Not relevant for 6G in 2030 in the upper mid-band

From **Massive** to **Gigantic** Numbers

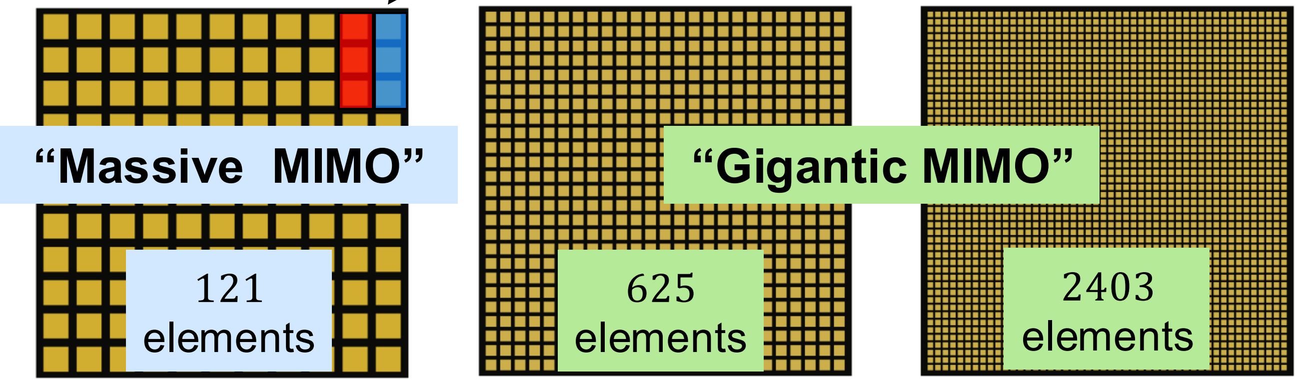
5G base station:

Example: 0.5×0.5 m arrays

Antenna ports

\# elements ≥ # antenna ports

- Fully digital transceivers
- Dual-polarized elements



3.5 GHz: 11×11 array

7.8 GHz: 25×25 array

15 GHz: 48×48 array

My **Definition** of **Gigantic MIMO**

Typical features

1. At least 256 antenna ports (M)
2. Operating in upper mid-band
3. Half-wavelength spacing
4. Fully digital beamforming
5. Spatial multiplexing of K streams
   (many users, a few per user)
6. Retain: $M \gg K$

768 element array
Ericsson booth
EuCAP 2025

Not needed

a) Rate-splitting multiple access (RSMA)

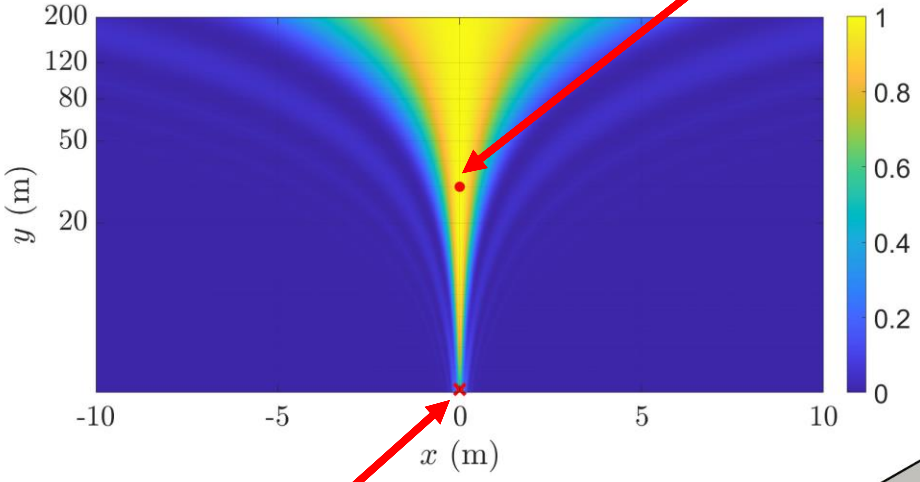
b) Movable antennas

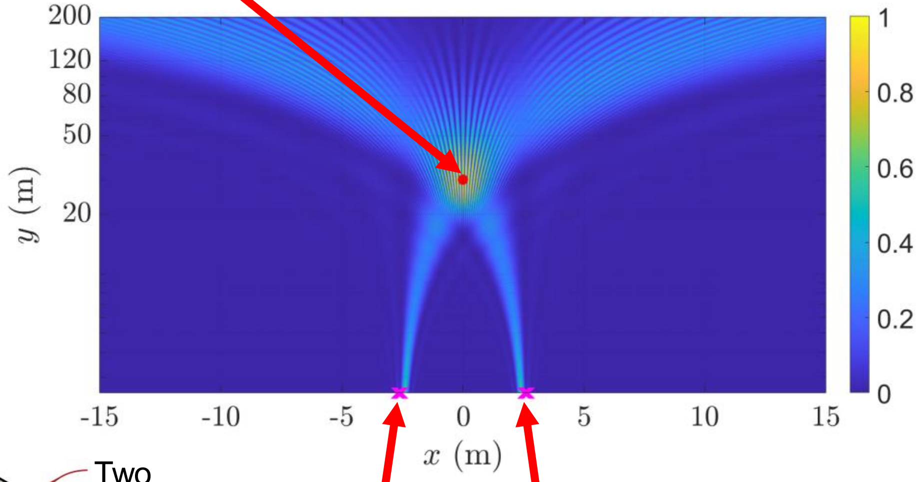
c) mmWave or THz bands

Can We Do Beamfocusing with Gigantic MIMO?

Yes, with multiple panels

Receiver at 30 m

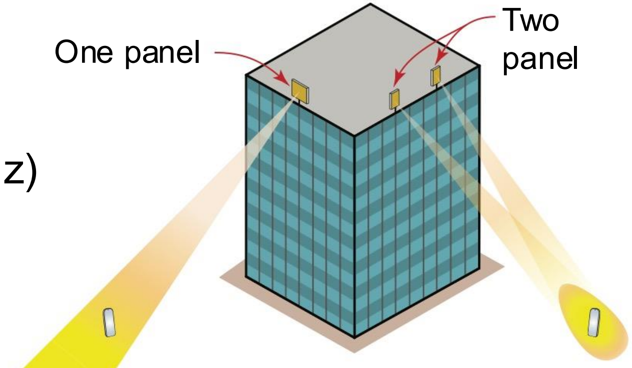




Uniform linear array (ULA)

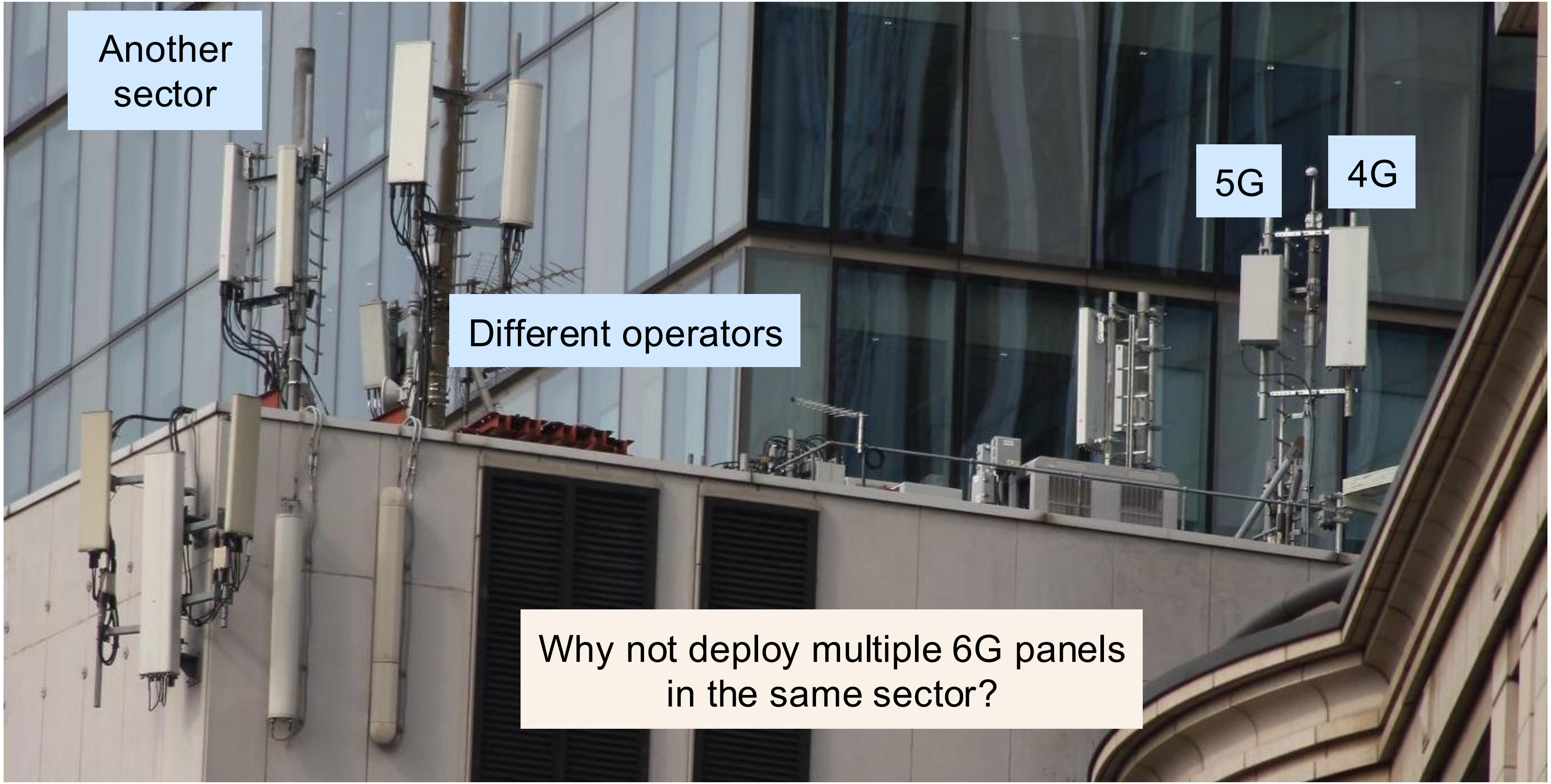
48 antennas (0.5 m at 15 GHz)

Fraunhofer distance: 23 m



Two 24-antenna ULAs, 5 m apart

A Practical Deployments in London

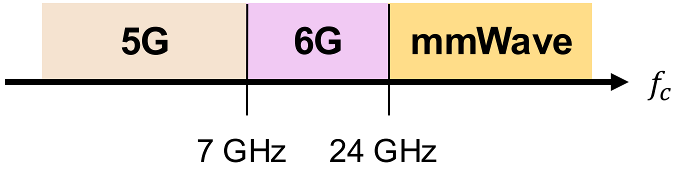


Picture by Peter Clarke, https://pedroc.co.uk/

Gigantic MIMO: Next-Generation Antenna Technology!

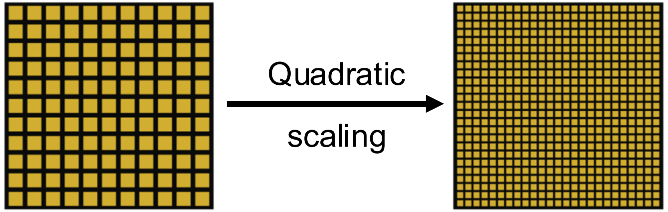
Frequency band: Upper mid-band (7-24 GHz)

- Mediocre bandwidth availability
- Many more antennas in the same form factor



"Gigantic MIMO"

- At least 256 antenna ports
- Capacity grows as: $\text{bandwidth} \times \text{antennas}$

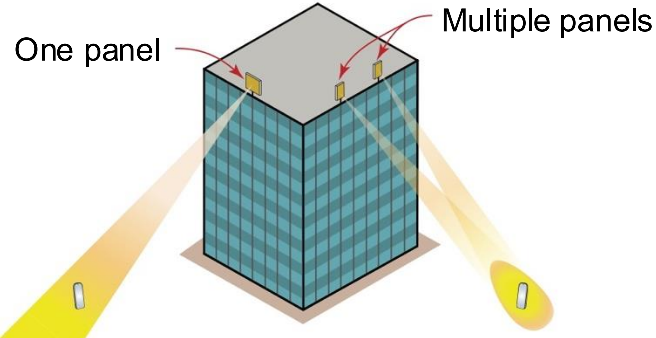


Enabling 6G Performance in the Upper Mid-Band by Transitioning From Massive to Gigantic MIMO

EMIL BJÖRNSON[1] (Fellow, IEEE), FERDI KARA[1,2] (Senior Member, IEEE),
NIKOLAOS KOLOMVAKIS[1] (Member, IEEE), ALVA KOSASIH[3] (Member, IEEE),
PARISA RAMEZANI[1] (Member, IEEE), AND MURAT BABEK SALMAN[1] (Member, IEEE)

Near-field effects

- Requires multi-panel deployments
- Precise beamfocusing for interference control
- Enabler for bistatic sensing



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ABSTRACT The initial 6G networks will likely operate in the upper mid-band (7-24 GHz), which has decent propagation conditions but underwhelming new spectrum availability. In this paper, we explore whether we can anyway reach the ambitious 6G performance goals by evolving the multiple-input multiple-output (MIMO) technology from massive in 5G to gigantic in 6G. We describe how many antennas are needed to reach the envisioned 6G peak user rates, how many can realistically be deployed in practical radio equipment, and what the practical spatial degrees-of-freedom might become. We further suggest a new deployment strategy that enables the utilization of radiative near-field effects in these bands for precise beamfocusing, localization, and sensing from a single base station site. Finally, we identify open research and standardization challenges that must be overcome to efficiently use gigantic MIMO dimensions in 6G from hardware, cost, and algorithmic perspectives.